Course Code: PH7302E **Session:** Winter Semester 2026–27
Course Title: Condensed Matter Physics **Duration:** 3 Hours
Max. Marks: 50 **Weightage:** Pre-Midterm 30% (15 M) / Post-Midterm 70% (35 M)

Instructions to Candidates:

1. Answer **ALL** questions. Total marks: **50**.
2. The examination is structured into two parts:
   - **PART A (Questions 1–2):** Pre-Midterm Syllabus **[15 Marks — 30%]**
   - **PART B (Questions 3–6):** Post-Midterm Syllabus **[35 Marks — 70%]**
3. Marks for each question and its sub-parts are indicated in the right margin.
4. State clearly any physical approximations and assumptions made.

Physical Constants: $h = 6.626\times10^{-34}$ J·s, $\hbar = 1.055\times10^{-34}$ J·s, $e = 1.602\times10^{-19}$ C, $m_e = 9.109\times10^{-31}$ kg, $k_B = 1.381 \times 10^{-23}$ J $\cdot$ K^{-1}, $N_A = 6.022 \times 10^{23}$ mol^{-1}, $\epsilon_0 = 8.854 \times 10^{-12}$ F $\cdot$ m^{-1}, $\mu_0 = 4\pi \times 10^{-7}$ H $\cdot$ m^{-1}, $\mu_B = 9.274 \times 10^{-24}$ J $\cdot$ T^{-1}, $c = 2.998 \times 10^{8}$ m $\cdot$ s^{-1}, 1 eV $= 1.602 \times 10^{-19}$ J, 1 amu $= 1.661 \times 10^{-27}$ kg.

PART A — PRE-MIDTERM SYLLABUS [15 MARKS]

Q1. [Crystal Structure, Reciprocal Lattice & X-ray Diffraction] [CO1]

(a) **Structure Factor and Systematic Extinctions:**

(i) Define the geometric structure factor F_{hkl} for a unit cell containing basis atoms at positions $\mathbf{r}_j$.

(ii) For a Face-Centered Cubic (FCC) lattice with identical atoms having atomic scattering factor f, evaluate F_{hkl} and deduce the systematic reflection selection rules. Prove that reflections with mixed Miller indices (some even, some odd) vanish identically. **[1, 3 Marks]**

(b) **Powder X-ray Diffraction Analysis:** A powder diffraction pattern of polycrystalline Gold (FCC structure, lattice parameter $a = 4.08$ Å) is recorded using monochromatic Cu K_α radiation ($\lambda = 1.5406$ Å).

(i) Determine the Miller indices (hkl) of the first two diffraction peaks observed at the lowest Bragg angles.

(ii) Calculate their corresponding interplanar spacings d_{hkl} (in Å) and the respective glancing Bragg angles θ (in degrees). **[1, 2 Marks]**

Q2. [Lattice Vibrations, Phonon Transport & Free Electron Gas] [CO2, CO3]

(a) **Thermal Conductivity and Phonon Scattering:** The lattice thermal conductivity in an insulator is expressed as $\kappa_{ph} = \frac{1}{3} C_v v \ell_{ph}$.

(i) Detail the physical mechanisms governing the temperature variation of $\kappa_{ph}(T)$ in three regimes: $T \to 0$ K, intermediate temperatures ($T \sim 0.1\,\Theta_D$), and high temperatures ($T \gg \Theta_D$).

(ii) Differentiate between Normal (N) and Umklapp (U) three-phonon scattering processes. With the help of a reciprocal space vector diagram, explain why Umklapp processes are indispensable for establishing a non-zero thermal resistance in pure dielectric crystals. **[3, 2 Marks]**

(b) **Wiedemann-Franz Relation and Electronic Thermal Conduction:**

(i) State the Wiedemann-Franz law. Using the Sommerfeld quantum theory of metals, write the analytical expression for the Lorenz number $L = \frac{\kappa_e}{\sigma T}$ and compute its theoretical numerical value.

(ii) Copper has an electrical conductivity $\sigma = 5.90 \times 10^7\ \Omega^{-1} \cdot \text{m}^{-1}$ at $T = 300$ K. Estimate its electronic thermal conductivity κ_e at this temperature. **[1, 2 Marks]**

PART B — POST-MIDTERM SYLLABUS [35 MARKS]

Q3. [Bloch's Theorem, Nearly Free Electron Model & Tight-Binding Theory] [CO4]

(a) **Nearly Free Electron (NFE) Model and Band Gap Formation:**

(i) State Bloch's theorem for an electron moving in a 1D crystal potential $V(x) = V(x + a)$.

(ii) Near the first Brillouin zone boundary ($k \approx \pm G/2 = \pm\pi/a$), the electron wavefunction is represented by a two-component plane-wave expansion:

$$\psi(x) = C_1 e^{ikx} + C_2 e^{i(k-G)x}, \quad \text{with } G = \frac{2\pi}{a}.$$

For a periodic potential $V(x) = 2V_1 \cos(Gx)$, set up the 2×2 secular determinant from the time-independent Schrödinger equation.

(iii) Solve this secular equation at the exact zone boundary $k = G/2$ and prove that an energy band gap of width $E_g = 2|V_1|$ opens between the two standing-wave states.

(iv) Sketch the resulting dispersion curve $E(k)$ across the first Brillouin zone in the *reduced zone scheme*, clearly labeling the band gap and energy extrema. **[1, 2, 2, 1 Marks]**

(b) **Tight-Binding Approximation (LCAO):**

(i) For a one-dimensional linear atomic chain with lattice constant a, consider an electron occupying tightly bound atomic s-orbitals. Assuming only nearest-neighbor overlap integral t and on-site energy ϵ_0, derive the energy dispersion relation:

$$E(k) = \epsilon_0 - 2t\cos(ka).$$

(ii) Find the total bandwidth W of this energy band.

(iii) Find the effective mass m^* at the band bottom ($k = 0$) and top ($k = \pi/a$). **[2, 1, 1 Marks]**

Q4. [Electron Dynamics in Periodic Lattices & Semiconductor Physics] [CO4]

(a) **Band Electron Dynamics, Negative Effective Mass, and Holes:**

(i) From the definition of group velocity $v_g = \frac{1}{\hbar}\frac{dE}{dk}$ and the acceleration caused by an applied electric field $\mathcal{E}$, derive the fundamental effective mass equation:

$$m^* = \hbar^2 \left(\frac{d^2E}{dk^2}\right)^{-1}.$$

(ii) Explain the physical significance of a negative effective mass near the upper edge of an energy band. Explain how an unfilled electron state in an otherwise full valence band is treated as a quasiparticle "hole" with charge $+e$ and positive mass $+|m^*|$.

(iii) Define *Bloch oscillations*. Why can they not be observed in typical bulk metals under ambient conditions? **[1, 2, 1 Marks]**

(b) **Semiconductor Statistics and Carrier Concentrations:**

(i) In a 3D intrinsic semiconductor with band gap $E_g = E_c - E_v$, derive the temperature-dependent intrinsic carrier concentration:

$$n_i(T) = \sqrt{N_c N_v}\, \exp\left(-\frac{E_g}{2k_B T}\right),$$

where $N_c = 2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2}$ and $N_v = 2\left(\frac{2\pi m_h^* k_B T}{h^2}\right)^{3/2}$.

(ii) Derive the expression for the intrinsic Fermi level E_{Fi} as a function of temperature:

$$E_{Fi} = \frac{E_c + E_v}{2} + \frac{3}{4}k_B T \ln\left(\frac{m_h^*}{m_e^*}\right).$$

(iii) Pure Silicon at 300 K has $E_g = 1.12$ eV, $m_e^* = 1.08\, m_e$, and $m_h^* = 0.56\, m_e$. Calculate N_c, N_v, and n_i (in m^{-3} and cm^{-3}). Find the energy shift of E_{Fi} from midgap. **[1, 1, 2 Marks]**

Q5. [Dielectric Properties, Local Field & Optical Responses of Solids] [CO5]

(a) **Local Electric Field and Clausius-Mossotti Relation:**

(i) Differentiate between the macroscopic average electric field $\mathbf{E}$ and the microscopic local field $\mathbf{E}_{loc}$ acting at a lattice site in a dielectric.

(ii) By evaluating the polarization charge on a fictitious Lorentz spherical cavity in a cubic dielectric, show that:

$$\mathbf{E}_{loc} = \mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0}.$$

(iii) Combining the microscopic dipole response $\mathbf{P} = N\alpha\mathbf{E}_{loc}$ with the macroscopic constitutive relation $\mathbf{D} = \epsilon_0\epsilon_r\mathbf{E} = \epsilon_0\mathbf{E} + \mathbf{P}$, derive the *Clausius-Mossotti equation*:

$$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}.$$

(iv) Explain what is meant by the "polarization catastrophe" and how this instability triggers the ferroelectric state with spontaneous polarization. **[1, 1, 2, 1 Marks]**

(b) **Optical Properties and Plasma Oscillations of Conduction Electrons:**

(i) Neglecting collision damping, set up the equation of motion for a free conduction electron in an oscillating electric field $\mathbf{E}(t) = \mathbf{E}_0 e^{-i\omega t}$. Derive the frequency-dependent dielectric function of the electron gas:

$$\epsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2}, \quad \text{where } \omega_p = \sqrt{\frac{ne^2}{\epsilon_0 m_e}}.$$

(ii) Discuss the optical response of metals in terms of the complex refractive index $\tilde{n} = \sqrt{\epsilon(\omega)}$: explain why metals exhibit high reflectivity ($R \approx 1$) for $\omega < \omega_p$ and become transparent to electromagnetic radiation for $\omega > \omega_p$.

(iii) An alkali metal has a conduction electron density $n = 6.00 \times 10^{28}$ m^{-3}. Calculate its plasma frequency ω_p (in rad $\cdot$ s^{-1}), cyclic frequency ν_p (in Hz), and the corresponding ultraviolet cutoff wavelength λ_p (in nm). **[1, 1, 2 Marks]**

Q6. [Exchange Interaction, Ferromagnetism & Antiferromagnetism] [CO5]

(a) **Weiss Molecular Field Theory of Ferromagnetism:**

(i) State the physical premise of Weiss's molecular field hypothesis, writing the internal field as $B_{eff} = B_a + \lambda M$, where λ is the molecular field parameter.

(ii) Using the Curie law $\chi = M/B_{eff} = C/T$ for paramagnetism above the Curie temperature T_C, derive the *Curie-Weiss law*:

$$\chi = \frac{C}{T - \theta_C}, \quad \text{with } \theta_C = \lambda C.$$

(iii) In the absence of an applied magnetic field ($B_a = 0$), show analytically or graphically using the transcendental equation $M/M_s = \tanh\left(\frac{g\mu_B S B_{eff}}{k_B T}\right)$ that a stable spontaneous magnetization $M_s \neq 0$ appears only when $T < T_C$. **[1, 1, 2 Marks]**

(b) **Heisenberg Exchange Interaction and Antiferromagnetism:**

(i) Explain why classical magnetic dipole-dipole interactions fail by several orders of magnitude to explain the observed transition temperatures ($T_C \sim 1000$ K) of ferromagnets.

(ii) Explain the quantum origin of the Heisenberg exchange Hamiltonian:

$$\hat{H}_{ex} = -2\sum_{i<j} J_{ij}\,\mathbf{S}_i \cdot \mathbf{S}_j,$$

clarifying how Coulomb repulsion combined with the Pauli exclusion principle dictates the sign of the exchange integral J. Distinguish between ferromagnetic ($J > 0$) and antiferromagnetic ($J < 0$) coupling.

(iii) Sketch the temperature dependence of magnetic susceptibility $\chi(T)$ for an antiferromagnet, clearly highlighting the behavior above and below the Néel temperature T_N for fields applied parallel ($\chi_\parallel$) and perpendicular ($\chi_\perp$) to the spin axis. **[1, 2, 1 Marks]**

Question-wise Course Outcome (CO) Mapping and Bloom's Taxonomy Level:

Question No.	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Total**	**Wt.**
Syllabus Domain	Pre-Mid	Pre-Mid	Post-Mid	Post-Mid	Post-Mid	Post-Mid	—	—
Course Outcome	CO1	CO2, CO3	CO4	CO4	CO5	CO5	CO1–CO5	—
Bloom's Level*	Level 2–3	Level 2–3	Level 3–4	Level 3–4	Level 3–4	Level 3–4	—	—
Marks Allocated	**7**	**8**	**10**	**8**	**9**	**8**	**50**	**100%**

*Bloom's Revised Taxonomy: Level 1 = Remembering; Level 2 = Understanding; Level 3 = Applying; Level 4 = Analysing; Level 5 = Evaluating.

Course Outcomes (PH7302E): CO1: Differentiate lattice types, reciprocal lattice and crystal diffraction. **CO2:** Distinguish crystal excitations and explain specific heat. **CO3:** Apply free electron model and interpret metallic properties. **CO4:** Explain energy bands and electrical properties of materials. **CO5:** Explain dielectric, optical, and magnetic properties of materials.

***** END OF QUESTION PAPER — ALL THE BEST *****